

Final exam test bank

1. Dormant, thick-walled resting cells or cyanobacteria that are resistant to desiccation are called:
 - ☐ a. hormogonia
 - ☐ b. akinetes
 - ☐ c. baeocytes
 - ☐ d. thylakoids
2. The part of the spirochete that houses the cytoplasm and the nucleoid of spirochetes is called the _____ cylinder.
 - ☐ a. mesoplasmic
 - ☐ b. cytoplasmic
 - ☐ c. ectoplasmic
 - ☐ d. protoplasmic
3. Which of the following is true of the actinomycetes?
 - ☐ a. they divide by budding.
 - ☐ b. like other filamentous bacteria, the hyphae do not undergo fragmentation.
 - ☐ c. some are photosynthetic.
 - ☐ d. they do not use hyphal fragmentation as a growth strategy.
4. The chlamydias are:
 - ☐ a. motile due to periplasmic flagella.
 - ☐ b. obligate intracellular parasites.
 - ☐ c. gram-positive spore-forming rods.
 - ☐ d. photosynthetic.
5. The host on or in which the parasitic organisms either attain sexual maturity or reproduce is the:
 - ☐ a. final host.
 - ☐ b. reservoir host.
 - ☐ c. transfer host.
 - ☐ d. intermediate host.
6. Which of the following is not one of the groups of streptococci?
 - ☐ a. the pyogenic streptococci



- o b. the anaerobic streptococci
 - o c. the oral streptococci
 - o d. the dermal streptococci
7. An organism other than a human that is infected with a parasitic organism that can also infect humans is called a:
- o a. transfer host
 - o b. reservoir host
 - o c. intermediate host
 - o d. endoparasite
8. All the following are microbial products involved in bacterial dissemination in mammalian cells except:
- o a. fimbriae
 - o b. coagulase
 - o c. hydrogen peroxide
 - o d. protein A
9. Which of the following is NOT true regarding bacterial virulence factors involved in bacterial pathogen invasion & dissemination?
- o a. *S. aureus* produces coagulase which coagulates the fibrinogen in plasma.
 - o b. *Clostridium* spp. produce collagenase to break down collagen to assist the spread of the pathogen.
 - o c. Staphylococci produce hyaluronidase to hydrolyze hyaluronic acid, a constituent of the extracellular matrix.
 - o d. Staphylococci produce hemolysins important in lysing leukocytes.
10. The sac fungi are distinguished from all other groups of fungi because their sexual spores are:
- o a. borne on basidia.
 - o b. motile.
 - o c. zygospores.
 - o d. inside an ascus.

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	Q10
b	d	d	b	a	d	b	c	d	d



11. Zygomycetes live on:

- ☐ a. fruits and vegetables.
- ☐ b. bread
- ☐ c. all of the choices
- ☐ d. organic matter in the soil.

12. Which of the following genera grow symbiotically within root nodules of legumes as nitrogen-fixing bacteroids?

- ☐ a. Rhizobium
- ☐ b. Nitrosomonas
- ☐ c. Agrobacterium
- ☐ d. Bacteroides

13. The genus *Desulfovibrio*:

- ☐ a. produces H₂S.
- ☐ b. all of the choices
- ☐ c. uses sulfate as a terminal electron acceptor during anaerobic respiration.
- ☐ d. is a strict anaerobe.

14. Fungi:

- ☐ a. produce chlorophyll under aerobic conditions.
- ☐ b. are prokaryotes or eukaryotes, depending on the species.
- ☐ c. are comprised of eukaryotic cells.
- ☐ d. produce chlorophyll under aerobic conditions and are prokaryotes or eukaryotes, depending on the species.

15. Members of the genus *Leuconostoc* are important in the production of:

- ☐ a. sauerkraut.
- ☐ b. cheese.
- ☐ c. dextrans.
- ☐ d. all of these

16. Which of the following is not useful in classifying actinomycetes?

- ☐ a. mycelial morphology
- ☐ b. color of the sporangia



- o c. sporangial morphology
- o d. ability to ferment carbohydrates

17. Members of the genus *Micrococcus* are usually:

- o a. facultatively anaerobic.
- o b. anaerobic.
- o c. microaerophilic.
- o d. aerobic.

18. Actinomycete spores formed within a special structure are called:

- o a. sporangiospores.
- o b. myxospores.
- o c. endospores.
- o d. conidiospores.

19. Actinomycetes:

- o a. form heat-resistant spores.
- o b. form sexual spores, that is, they are used for sexual reproduction.
- o c. do not form spores.
- o d. form asexual spores.

20. *Beggiatoa* are:

- o a. obligate anaerobes.
- o b. facultative anaerobes.
- o c. obligate aerobes.
- o d. microaerophiles.

Q11	Q12	Q13	Q14	Q15	Q16	Q17	Q18	Q19	Q20
d	a	b	c	d	d	d	a	d	c

21. Which of the following diseases is caused by *Clostridium perfringens*?

- o a. pneumonia
- o b. fire blight disease of pears
- o c. tuberculosis



- ☐ d. gas gangrene

22. Which of the following is true of *Agrobacterium tumefaciens*?

- ☐ a. it causes crown gall disease when it carries a tumor-inducing (TI) plasmid
- ☐ b. it may be a member of the same species as *Rhizobium*.
- ☐ c. it is not capable of nitrogen fixation.
- ☐ d. all of the choices are true.

23. Actinomycetes:

- ☐ a. do not form spores.
- ☐ b. form asexual spores.
- ☐ c. form sexual spores, that is, they are used for sexual reproduction.
- ☐ d. form heat-resistant spores.

24. Which of the following is not a characteristic of an endotoxin?

- ☐ a. neurotoxic
- ☐ b. heat stable
- ☐ c. weakly immunogenic
- ☐ d. moderate toxicity

25. Fungi:

- ☐ a. are comprised of eukaryotic cells.
- ☐ b. are prokaryotes or eukaryotes, depending on the species.
- ☐ c. produce chlorophyll under aerobic conditions and are prokaryotes or eukaryotes, depending on the species.
- ☐ d. produce chlorophyll under aerobic conditions.

26. Streptococcal sore throat (strep throat) is caused by:

- ☐ a. *Streptococcus mutans*.
- ☐ b. *Streptococcus pneumoniae*.
- ☐ c. *Streptococcus pyogenes*.
- ☐ d. *Streptococcus trachius*.

27. Which of the following antibiotics are produced by members of the genus *Bacillus*?

- ☐ a. all of these
- ☐ b. polymyxin
- ☐ c. bacitracin



- o d. gramicidin

28. The spirochetes include the causative agents for:

- o a. plague.
- o b. gonorrhea.
- o c. Lyme disease.
- o d. whooping cough.

29. *Helicobacter pylori* is responsible for:

- o a. cholera
- o b. dysentery.
- o c. peptic ulcer disease.
- o d. pneumonia

30. The fruiting body of a fungus produces:

- o a. zygotes.
- o b. rhizoids.
- o c. gametes.
- o d. spores.

Q21	Q22	Q23	Q24	Q25	Q26	Q27	Q28	Q29	Q30
d	d	b	a	a	c	a	c	c	d

31. Which of the following is NOT true regarding bacterial virulence factors involved in bacterial pathogen invasion & dissemination?

- o a. *S. aureus* produces coagulase which coagulates the fibrinogen in plasma.
- o b. *Clostridium* spp. produce collagenase to break down collagen to assist the spread of the pathogen.
- o c. Staphylococci produce hyaluronidase to hydrolyze hyaluronic acid, a constituent of the extracellular matrix.
- o d. Staphylococci produce hemolysins important in lysing leukocytes.

32. The body or vegetative structure of a fungus is called a:

- o a. mold



- ☐ b. yeast
 - ☐ c. thallus
 - ☐ d. hypha
- 33. Which of the following is NOT one of the groups of streptococci?
 - ☐ a. the dermal streptococci
 - ☐ b. the pyogenic streptococci
 - ☐ c. the anaerobic streptococci
 - ☐ d. the oral streptococci
- 34. In which of the following regions in the host bacterium are you most likely to find Bdellovibrio?
 - ☐ a. the periplasmic space
 - ☐ b. the nucleoid
 - ☐ c. the cell nucleus
 - ☐ d. the cytoplasm
- 35. Which of the following accumulates sulfur granules inside the cell?
 - ☐ a. green sulfur bacteria
 - ☐ b. cyanobacteria
 - ☐ c. purple sulfur bacteria
 - ☐ d. none of the above
- 36. The host on or in which the parasitic organisms either attain sexual maturity or reproduce is the:
 - ☐ a. intermediate host.
 - ☐ b. reservoir host.
 - ☐ c. final host.
 - ☐ d. transfer host.
- 37. Fungi are found primarily in which of the following environments?
 - ☐ a. freshwater
 - ☐ b. marine
 - ☐ c. terrestrial
 - ☐ d. arboreal
- 38. Coxiella burnetii reproduces in the _____ of a eukaryotic host cell.
 - ☐ a. nucleus
 - ☐ b. endoplasmic reticulum



- o c. cytoplasm
- o d. phagolysosome

39. An infection caused by pus-producing organisms is called:

- o a. toxemic
- o b. systemic
- o c. pyogenic
- o d. septic

40. In which of the following situations did disease develop due to vertical transmission of the pathogen?

- o a. A traveler develops gastroenteritis after drinking water contaminated with Escherichia coli bacteria.
- o b. A dog develops rabies after being scratched during a fight with a cat that carries the virus.
- o c. A hospitalized patient develops MRSA bacteria in a wound following surgery.
- o d. A fetus develops listeriosis after unpasteurized cheese infected with Listeria bacteria is consumed by the mother during pregnancy.

Q31	Q32	Q33	Q34	Q35	Q36	Q37	Q38	Q39	Q40
d	c	a	a	c	c	c	d	c	d

41. Endotoxins include which of the following?

- o a. botulinum toxin
- o b. tetanus toxin
- o c. diphtheria toxin
- o d. lipopolysaccharide

42. Which of the following is used in the production of buttermilk and cheese?

- o a. Streptococcus pneumoniae
- o b. S. faecalis
- o c. Lactococcus lactis
- o d. S. mutans

43. Which of the following is true of the actinomycetes?

- o a. like other filamentous bacteria, the hyphae do not undergo fragmentation.



- ☐ b. some are photosynthetic.
 - ☐ c. they divide by budding.
 - ☐ d. they do not use hyphal fragmentation as a growth strategy.
- 44. Which of the following genera grow symbiotically within root nodules of legumes as nitrogen-fixing bacteroids?
 - ☐ a. Bacteroides
 - ☐ b. Agrobacterium
 - ☐ c. Rhizobium
 - ☐ d. Nitrosomonas
- 45. Which of the following contain both bacteriochlorophyll a and bacteriochlorophyll b?
 - ☐ a. green sulfur bacteria
 - ☐ b. purple sulfur bacteria
 - ☐ c. cyanobacteria
 - ☐ d. Prochlorales
- 46. Which of the following statements describe the genus Caryophanon?
 - ☐ a. all of the above
 - ☐ b. its cells have a distinctive disc shape and are joined together to form rods.
 - ☐ c. it is strictly aerobic and catalase positive.
 - ☐ d. its normal habitat is in cow dung.
- 47. Beggiatoa are:
 - ☐ a. microaerophiles.
 - ☐ b. obligate aerobes.
 - ☐ c. obligate anaerobes.
 - ☐ d. facultative anaerobes.
- 48. Members of the genus Thermoactinomyces:
 - ☐ a. are commonly found in damp haystacks and compost piles.
 - ☐ b. are thermophiles.
 - ☐ c. all of these
 - ☐ d. produce endospores.
- 49. The bacteria that form colonies like fried eggs on agar are:
 - ☐ a. Clostridium tetanus



- o b. *Lactobacillus lactis*
- o c. *Mycoplasma pneumoniae*
- o d. *Toxoplasma gondii*

50. *Campylobacter jejuni* is the cause of:

- o a. wilt disease in plants.
- o b. meningitis.
- o c. abortion in sheep and enteritis diarrhea in humans.
- o d. fowl cholera.

Q41	Q42	Q43	Q44	Q45	Q46	Q47	Q48	Q49	Q50
d	c	d	c	b	a	b	c	c	c

51. Which of the following is NOT true regarding bacterial virulence factors involved in bacterial pathogen invasion & dissemination?

- a. *Clostridium* spp. produce collagenase to break down collagen to assist the spread of the pathogen.
- b. Staphylococci produce hemolysins important in lysing leukocytes.
- c. *S. aureus* produces coagulase which coagulates the fibrinogen in plasma.
- d. Staphylococci produce hyaluronidase to hydrolyze hyaluronic acid, a constituent of the extracellular matrix.

52. Which of the following is not found among the zygomycetes, ascomycetes, and basidiomycetes?

- a. flagellated cells
- b. fusion of gametes
- c. saprophytic nutrition
- d. septate hyphae

53. Which of the following groups of photosynthetic bacteria are capable of oxygenic photosynthesis?

- a. the green bacteria
- b. the purple bacteria
- c. the cyanobacteria
- d. all of the above



54. The conidiospores of streptomycetes are:

- a. motile by means of cilia.
- b. nonmotile.
- c. motile by means of pseudopodia.
- d. motile by means of flagella.

55. The complex of periplasmic flagella that mediates the movement of spirochetes is referred to as the:

- a. peritrichal axon.
- b. flagellar complex.
- c. axial filament.
- d. axillary filament.

56. Abscesses and boils are often associated with:

- a. *Lactococcus lactis*.
- b. *Staphylococcus aureus*.
- c. *Streptococcus mutans*.
- d. *Streptococcus pneumoniae*.

Q50	Q51	Q52	Q53	Q54	Q55	Q56
c	b	a	c	b	c	b

57. The purple non-sulfur bacteria:

- a. utilize organic molecules as a source of electrons for photosynthesis.
- b. typically can grow aerobically in the dark as chemoorganoheterotrophs.
- c. grow anaerobically as photoorganoheterotrophs.
- d. all of the above

58. Which of the following is not a mode of action associated with endotoxin?

- a. induction of paralysis



- b. production of tissue necrosis factor Alpha
- c. generation of signals resulting in fever
- d. activation of the coagulation cascade

59. The mycoplasma is naturally resistant to penicillin due to:

- a. it produces a high amount of beta-lactamase which prevents the penicillin from entering the cell wall.
- b. it has no peptidoglycan in its cell wall.
- c. it has no cell wall.
- d. it produces a high amount of penicillinase which breaks down penicillin.

60. Beggiatoa are:

- a. obligate aerobes.
- b. obligate anaerobes.
- c. microaerophiles.
- d. facultative anaerobes.

61. In fungi, spores produced by budding are called:

- a. sporangiospores.
- b. blastospores.
- c. conidiospores.
- d. arthrospores.

62. A genus in the family Enterobacteriaceae that is a major pathogen of crop plants is:

- a. Proteus
- b. Erwinia
- c. Shigella
- d. Yersinia

63. In mushrooms, the sexual spores are produced in:

- a. ascus.



- b. sporophyte.
- c. basidium.
- d. sporangiophore.

64. Members of the genus *Thermoactinomyces*:

- a. are commonly found in damp haystacks and compost piles.
- b. are thermophiles.
- c. all of these
- d. produce endospores.

65. All the following are microbial products involved in bacterial dissemination in mammalian cells except:

- a. protein A
- b. coagulase
- c. fimbriae
- d. hydrogen peroxide

66. *Pasteurella multocida* causes:

- a. fowl cholera
- b. none of these
- c. abortion in sheep
- d. pneumonia in cattle and sheep

Q57	Q58	Q59	Q60	Q61	Q62	Q63	Q64	Q65	Q66
d	a	c	a	b	b	c	c	d	a

67. The fruiting body of a fungus produces:

- a. rhizoids.
- b. gametes.
- c. spores.
- d. zygotes.



68. In mushrooms, the sexual spores are produced in:

- a. ascus.
- b. sporophyte.
- c. basidium.
- d. sporangiophore.

69. Members of the genus *Micrococcus* usually exist as:

- a. all of these
- b. tetrads.
- c. pairs.
- d. irregular clusters.

70. Which of the following is pyogenic (pus-forming)?

- a. *Streptococcus mutans*
- b. *Streptococcus pneumoniae*
- c. *Streptococcus lactis*
- d. *Streptococcus faecalis*

71. The genus *Desulfovibrio*:

- a. all of the choices
- b. produces H₂S.
- c. is a strict anaerobe.
- d. uses sulfate as a terminal electron acceptor during anaerobic respiration.

72. All the following are microbial products involved in bacterial dissemination in mammalian cells except:

- a. fimbriae
- b. hydrogen peroxide
- c. coagulase
- d. protein A



73. *Coxiella burnetii* reproduces in the _____ of a eukaryotic host cell.

- a. nucleus
- b. phagolysosome
- c. cytoplasm
- d. endoplasmic reticulum

74. Which of the following is NOT one of the groups of streptococci?

- a. the dermal streptococci
- b. the pyogenic streptococci
- c. the anaerobic streptococci
- d. the oral streptococci

75. Which of the following is NOT true regarding bacterial virulence factors involved in bacterial pathogen invasion & dissemination?

- a. *Clostridium* spp. produce collagenase to break down collagen to assist the spread of the pathogen.
- b. Staphylococci produce hemolysins important in lysing leukocytes.
- c. *S. aureus* produces coagulase which coagulates the fibrinogen in plasma.
- d. Staphylococci produce hyaluronidase to hydrolyze hyaluronic acid, a constituent of the extracellular matrix.

76. *Thermoactinomyces vulgaris* is the causative agent for:

- a. primary atypical pneumonia.
- b. vaginitis.
- c. cystic fibrosis.
- d. farmer's lung disease.

Q67	Q68	Q69	Q70	Q71	Q72	Q73	Q74	Q75	Q76
c	c	a	b	a	b	b	a	b	d



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- c. coagulase
- d. protein A

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82. Which of the following is pyogenic (pus-forming)?

- a. *Streptococcus mutans*
- b. *Streptococcus pneumoniae*
- c. *Streptococcus lactis*
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83. Members of the genus *Micrococcus* usually exist as:

- a. all of these
- b. tetrads.
- c. pairs.
- d. irregular clusters.

84. In mushrooms, the sexual spores are produced in:

- a. ascus.
- b. sporophyte.
- c. basidium.
- d. sporangiophore.

85. The fruiting body of a fungus produces:

- a. rhizoids.



- b. gametes.
- c. spores.
- d. zygotes.

Q76	Q77	Q78	Q79	Q80	Q81	Q82	Q83	Q84	Q85
d	b	a	b	b	a	b	a	c	c

86. Which of the following is NOT true regarding bacterial virulence factors involved in bacterial pathogen invasion & dissemination?

- ☐ a. Clostridium spp. produce collagenase to break down collagen to assist the spread of the pathogen.
- ☐ b. Staphylococci produce hemolysins important in lysing leukocytes.
- ☐ c. S. aureus produces coagulase which coagulates the fibrinogen in plasma.
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- ☐ d. sporangiophore.

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- ☐ c. *S. aureus* produces coagulase which coagulates the fibrinogen in plasma.



- o d. Staphylococci produce hyaluronidase to hydrolyze hyaluronic acid, a constituent of the extracellular matrix.

Q86	Q87	Q88	Q89	Q90	Q91	Q92	Q93	Q94	Q95
b	a	b	B	a	b	a	c	c	b

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- a. Streptococcus mutans
- b. Streptococcus pneumoniae
- c. Streptococcus lactis
- d. Streptococcus faecalis

Q96	Q97	Q98	Q99	Q100
a	b	b	a	b

